Steel Tower Risk Visualization
Project Report

Date: March 2026
Version: 1.0 — Initial Release

Executive Summary

What this project delivers

A map showing all 113,509 steel transmission towers across the United States. Each tower is color-coded based on historical outage records and weather conditions in its area. Field service teams can use this map to decide which towers need attention first.

Osmose manages a large field service team responsible for inspecting and maintaining steel transmission towers. The core challenge was a familiar one in the utility industry: with over 100,000 towers spread across the entire country, where do you send your teams first?

This project answers that question by combining four publicly available datasets, connecting them to every individual tower, and producing a simple visual — green, amber, or red — that any field manager can act on immediately, without needing to understand the data behind it.

113,509 Towers mapped	**4** Datasets combined	**2015–2023** Historical data range
22,865 High exposure towers (red)	**34,165** Moderate exposure (amber)	**56,479** Low exposure towers (green)

Data sources: HIFLD (hifld-geoplatform.hub.arcgis.com) | NOAA (ncdc.noaa.gov) | DOE (oe.netl.doe.gov) | NARR (psl.noaa.gov)

1. User Requirements

Osmose defined the project scope in four clear phases. Each phase builds on the previous one, moving from simply showing towers on a map to providing actionable field service guidance.

1.1 What Osmose asked for

Phase	Goal	Description
1	**Map towers**	Place all steel transmission towers on an interactive map of the United States so the field team can see where every tower is located.
2	**Add outages**	Overlay historical outage data on the map so teams can see where major grid failures have occurred in the past.
3	**Correlate outages**	Assign a visibility level (risk level) to towers based on how severe nearby outages were — measured by the number of customers affected.
4	**Integrate weather**	Add weather overlays including storm tracks, wind data, and snow depth so teams can see which towers sit in historically harsh weather zones.

1.2 What the tool is and is not

Important: this is a visibility tool, not a prediction tool

This map shows historical exposure conditions — where severe outages and storms have occurred near each tower. It does not predict which towers will fail in the future. A red tower means the area has a history of severe grid events. The field team decides what to do based on that information.

1.3 Why these matters for Osmose

- Osmose operates a large field service team across the US — prioritising their work is critical to efficiency
- Without data-driven visibility, inspection decisions rely on guesswork or outdated schedules
- This tool gives field managers a real-time, data-backed view of which areas deserve attention first
- It allows Osmose to demonstrate monitoring capability to utility clients and justify their field service investment

Data sources: HIFLD (hifld-geoplatform.hub.arcgis.com) | NOAA (ncdc.noaa.gov) | DOE (oe.netl.doe.gov) | NARR (psl.noaa.gov)

2. Data Sources

Four publicly available datasets were used. All data is free and comes from official US government or open-source repositories. No proprietary data was required for this project.

2.1 Overview of all four sources

Data Source	What It Contains	Where to Get It	How Used
HIFLD — Electric Transmission Lines	Location, voltage, owner and status of every major transmission line in the US. Used to derive tower positions.	hifld-geoplatform.hub.arcgis.com — free download as Shapefile or GeoJSON	Primary source of all 113,509 tower locations
NARR — North American Regional Reanalysis	Historical weather measurements including wind speed, temperature, precipitation and snow depth at grid points across North America.	psl.noaa.gov/data/gridded/data.narr.html — free NetCDF download	Weather conditions at each tower: max wind, min temp, snow depth
NOAA Storm Events Database	169,127 records of severe weather events (tornadoes, high wind, ice storms, etc.) from 2015 to 2023 with location coordinates.	ncdc.noaa.gov/stormevents/ftp.jsp — free annual CSV files	Storm history within 50 km of each tower
DOE OE-417 — Electric Emergency Reports	1,544 records of major grid disturbances filed by utilities to the US Department of Energy, including customers affected and MW lost. Extracted from annual PDF reports.	oe.netl.doe.gov/OE417_annual_summary.aspx — free PDF download	Outage severity within 100 km of each tower

2.2 Detailed source descriptions

Source 1 — HIFLD Transmission Lines

The Homeland Infrastructure Foundation-Level Data (HIFLD) programme maintains a national database of electric power infrastructure. Their transmission lines dataset covers the physical routes of high-voltage lines across the entire continental US, Hawaii and Alaska.

- Link: https://hifld-geoplatform.hub.arcgis.com/datasets/electric-power-transmission-lines
- Format: Shapefile or GeoJSON — free download, no account required
- Key fields used: geometry (line route), VOLTAGE (kV), VOLT_CLASS, OWNER, STATUS
- Tower positions derived by placing a point every 300 metres along each line — standard industry approximation since tower spacing on steel lattice lines ranges from 200m to 400m depending on voltage

- Result: 113,509 steel tower points with coordinates, owner and voltage class

1		@id	geometry	material	structure	design	distance_to_line	VOLTAGE	VOLT_CLASS	OWNER	STATUS	longitude	latitude
2	0	node/33349175	POINT (-93.3607767 44.9203384)	steel	lattice	barrel	4.06E-05	115	100-161	NORTHERN STATES POWER CO - MINNESOTA	IN SERVICE	-93.3607767	44.9203384
3	1	node/33351347	POINT (-93.3608119 44.9187157)	steel	lattice	barrel	3.29E-05	115	100-161	NORTHERN STATES POWER CO - MINNESOTA	IN SERVICE	-93.3608119	44.9187157
4	2	node/33370544	POINT (-93.3609065 44.9102856)	steel	lattice	barrel	3.30E-05	115	100-161	NORTHERN STATES POWER CO - MINNESOTA	IN SERVICE	-93.3609065	44.9102856
5	3	node/33371489	POINT (-93.3608931 44.9116597)	steel	lattice	barrel	2.82E-05	115	100-161	NORTHERN STATES POWER CO - MINNESOTA	IN SERVICE	-93.3608931	44.9116597
6	4	node/33524407	POINT (-93.3608387 44.9170403)	steel	lattice	barrel	3.39E-05	115	100-161	NORTHERN STATES POWER CO - MINNESOTA	IN SERVICE	-93.3608387	44.9170403
7	5	node/34397828	POINT (-93.3608619 44.9147955)	steel	lattice	barrel	3.13E-05	115	100-161	NORTHERN STATES POWER CO - MINNESOTA	IN SERVICE	-93.3608619	44.9147955
8	6	node/37022017	POINT (-76.7858858 39.228302)	steel	lattice	barrel	1.33E-06	230	220-287	BALTIMORE GAS & ELECTRIC CO	IN SERVICE	-76.7858858	39.228302
9	7	node/37022019	POINT (-76.7883096 39.2263948)	steel	lattice	barrel	1.48E-05	230	220-287	BALTIMORE GAS & ELECTRIC CO	IN SERVICE	-76.7883096	39.2263948
10	8	node/37022021	POINT (-76.7907287 39.2245009)	steel	lattice	barrel	3.09E-05	230	220-287	BALTIMORE GAS & ELECTRIC CO	IN SERVICE	-76.7907287	39.2245009
11	9	node/37022024	POINT (-76.795063 39.2211127)	steel	lattice	barrel	3.36E-05	230	220-287	BALTIMORE GAS & ELECTRIC CO	IN SERVICE	-76.795063	39.2211127
12	10	node/37022028	POINT (-76.7973907 39.2192972)	steel	lattice	barrel	1.50E-05	230	220-287	BALTIMORE GAS & ELECTRIC CO	IN SERVICE	-76.7973907	39.2192972
13	11	node/37022036	POINT (-76.8069983 39.2117885)	steel	lattice	barrel	2.26E-05	230	220-287	BALTIMORE GAS & ELECTRIC CO	IN SERVICE	-76.8069983	39.2117885
14	12	node/37022037	POINT (-76.809029 39.2101848)	steel	lattice	barrel	2.57E-06	230	220-287	BALTIMORE GAS & ELECTRIC CO	IN SERVICE	-76.809029	39.2101848
15	13	node/37022043	POINT (-76.810697 39.208871)	steel	lattice	barrel	2.45E-05	230	220-287	BALTIMORE GAS & ELECTRIC CO	IN SERVICE	-76.810697	39.208871

Source 2 — NARR Weather Reanalysis

The North American Regional Reanalysis (NARR) is produced by NOAA and provides historical weather data from 1979 to present at a 32km grid spacing across North America. Each tower was matched to its nearest NARR grid point to extract historical weather exposure.

- Link: https://psl.noaa.gov/data/gridded/data.narr.html
- Format: NetCDF files, downloadable by variable and date range — free
- Key fields extracted: wind_speed (m/s), temperature (K), precipitation (m/day), snow_depth (m)
- Aggregation: maximum wind speed, minimum temperature, total precipitation and maximum snow depth over the full historical record — peak exposure per tower

	A	B	C	D	E	F	G
1		@id	valid_time	wind_speed	temperature	precipitation	snow_depth
2	0	node/33349175	1/1/2016 12:00	4.013037	-8.438141	4.77E-07	0.0122509
3	1	node/33351347	1/1/2016 12:00	4.013037	-8.438141	4.77E-07	0.0122509
4	2	node/33370544	1/1/2016 12:00	4.013037	-8.438141	4.77E-07	0.0122509
5	3	node/33371489	1/1/2016 12:00	4.013037	-8.438141	4.77E-07	0.0122509
6	4	node/33524407	1/1/2016 12:00	4.013037	-8.438141	4.77E-07	0.0122509
7	5	node/34397828	1/1/2016 12:00	4.013037	-8.438141	4.77E-07	0.0122509
8	6	node/37022017	1/1/2016 12:00	2.6872458	3.776703	0	0
9	7	node/37022019	1/1/2016 12:00	2.6872458	3.776703	0	0
10	8	node/37022021	1/1/2016 12:00	2.6872458	3.776703	0	0
11	9	node/37022024	1/1/2016 12:00	2.6872458	3.776703	0	0
12	10	node/37022028	1/1/2016 12:00	2.6872458	3.776703	0	0
13	11	node/37022036	1/1/2016 12:00	2.6872458	3.776703	0	0

Source 3 — NOAA Storm Events Database

NOAA's National Centers for Environmental Information publishes a detailed record of significant weather events across the US. Each event record includes the event type, start and end date, location coordinates, damage caused and casualties. Annual CSV files were downloaded for 2015 through 2023.

- Link: https://ncdc.noaa.gov/stormevents/ftp.jsp — filename pattern: StormEvents_details-ftp_v1.0_d{year}_*.csv
- Format: CSV, one file per year — free download
- Event types retained: Thunderstorm Wind, High Wind, Tornado, Ice Storm, Winter Storm, Heavy Snow, Hurricane, Lightning, Strong Wind
- Key fields used: BEGIN_LAT, BEGIN_LON, EVENT_TYPE, DAMAGE_PROPERTY, DEATHS_DIRECT, INJURIES_DIRECT
- Coverage: 169,127 events with confirmed coordinates across 9 years

1	EVENT_ID	STATE	YEAR	MONTH_NA	BEGIN_DATE_TIME	END_DATE_TIME	EVENT_TYPE	MAGNIT	MAGNITU	DEATH	INJURIE	BEGIN_LAT	BEGIN_LON	END_LAT	END_LON	source_year	DAMAGE_PROPERTY_I	DAMAGE_CROPS_NUM
2	602745	IL	2015	September	9/18/2015 15:44	9/18/2015 15:53	Tornado			0	0	41.4164	-87.86	41.4266	-87.8119	2015	70000	0
3	594085	GEORGIA	2015	September	9/5/2015 14:18	9/5/2015 14:22	Thunderstorm Wind	50	EG	0	0	32.94	-82.2	32.94	-82.2	2015	2000	0
4	594086	SOUTH CAROLINA	2015	September	9/10/2015 16:28	9/10/2015 16:32	Thunderstorm Wind	50	EG	0	0	34.71	-80.72	34.71	-80.72	2015	14000	0
5	594083	GEORGIA	2015	September	9/4/2015 14:40	9/4/2015 14:44	Thunderstorm Wind	50	EG	0	0	32.94	-82.14	32.94	-82.14	2015	2000	0
6	594084	GEORGIA	2015	September	9/5/2015 14:17	9/5/2015 14:21	Thunderstorm Wind	50	EG	0	0	33.21	-82.15	33.21	-82.15	2015	3000	0
7	594087	SOUTH CAROLINA	2015	September	9/10/2015 16:28	9/10/2015 16:32	Thunderstorm Wind	50	EG	0	0	34.72	-80.75	34.72	-80.75	2015	12000	0
8	594088	SOUTH CAROLINA	2015	September	9/10/2015 18:30	9/10/2015 18:34	Thunderstorm Wind	50	EG	0	0	33.86	-80.05	33.86	-80.05	2015	3000	0
9	594089	SOUTH CAROLINA	2015	September	9/10/2015 17:00	9/10/2015 17:04	Thunderstorm Wind	68	MG	0	0	33.25	-81.65	33.25	-81.65	2015	0	0
10	594090	SOUTH CAROLINA	2015	September	9/10/2015 17:00	9/10/2015 17:04	Thunderstorm Wind	51	MG	0	0	33.21	-81.66	33.21	-81.66	2015	0	0
11	594091	SOUTH CAROLINA	2015	September	9/10/2015 18:55	9/10/2015 18:59	Thunderstorm Wind	55	EG	0	0	33.54	-80.2	33.54	-80.2	2015	6000	0
12	594092	SOUTH CAROLINA	2015	September	9/10/2015 18:55	9/10/2015 18:59	Thunderstorm Wind	55	EG	0	0	33.64	-80.07	33.64	-80.07	2015	4000	0
13	594094	GEORGIA	2015	September	9/10/2015 16:45	9/10/2015 16:49	Thunderstorm Wind	55	EG	0	0	32.97	-81.76	32.97	-81.76	2015	8000	0
14	594093	SOUTH CAROLINA	2015	September	9/10/2015 16:38	9/10/2015 16:42	Thunderstorm Wind	55	EG	0	0	34.72	-80.75	34.72	-80.75	2015	16000	0
15	595754	SOUTH CAROLINA	2015	September	9/21/2015 20:36	9/21/2015 20:40	Thunderstorm Wind	50	EG	0	0	33.9102	-80.6532	33.9102	-80.6532	2015	1000	0
16	594186	FLORIDA	2015	September	9/13/2015 6:53	9/13/2015 6:53	Thunderstorm Wind	40	EG	0	0	28.34	-82.62	28.34	-82.62	2015	1500	0
17	597596	NEW YORK	2015	September	9/9/2015 14:45	9/9/2015 14:45	Thunderstorm Wind	50	EG	0	0	44.36	-73.75	44.36	-73.75	2015	5000	0
18	597597	VERMONT	2015	September	9/9/2015 15:06	9/9/2015 15:06	Lightning			0	0	45	-72.4	45	-72.4	2015	1000	0

Source 4 — DOE OE-417 Emergency Reports

The US Department of Energy requires electric utilities to report major disturbances to the grid using Form OE-417. These reports are published annually as PDF documents. Each report lists events by month, including the utility name, area affected, type of disturbance, megawatts lost and customers affected.

- Link: https://oe.netl.doe.gov/OE417_annual_summary.aspx — annual PDF reports from 2015 to 2023
- Format: PDF — text extracted using pdfplumber Python library
- Challenge: PDFs have no structured data. Location is given as state and county names in narrative text, not coordinates. Custom text extraction was required.
- Location resolution: county names extracted from text, then matched to county centroid coordinates. Where county was not available, state centroid used. Where state was not available, NERC region centroid used.
- Result: 1,525 outage events with coordinates, out of 1,544 total records (98.7% coverage)
- Key fields extracted: event_start, mw_loss, customers_affected, primary_state, primary_county, utility, nerc_region, disturbance_type

1	year	month	event_start	event_end	duration	mw_loss	customers_	primary_state	primary_county	all_states	all_counties	utility	nerc_region	disturbance_type	source_file	lat	lng	location_sour
2	2022	1	1/2/2022 13:21	1/2/2022 17:38	4 Hours, 17 Minutes	0	1	California	Tuolumne	California	Tuolumne	Pacific Gas & Electric Co	WECC	Other	february2023.pdf	38.0276	-119.9548	county
3	2022	1	1/3/2022 1:00	1/3/2022 14:00	13 Hours, 0 Minutes	0	1	Georgia		Georgia, Alabama, Mississippi		Southern Company	SERC	Severe Weather	february2023.pdf	32.80868176	-83.5776956	state_centroi
4	2022	1	1/3/2022 5:00	1/4/2022 18:00	37 Hours, 0 Minutes	0	1	South Carolina		South Carolina, North Carolina		Dominion Energy South Carolina	SERC	Severe Weather	february2023.pdf	33.95771304	-80.9838	state_centroi
5	2022	1	1/3/2022 5:30	1/4/2022 12:30	31 Hours, 0 Minutes	0	142000	Massachusetts		Massachusetts		Duke Energy Carolinas	SERC	Severe Weather	february2023.pdf	42.11317143	-71.46901429	state_centroi
6	2022	1	1/5/2022 12:32	1/5/2022 12:45	0 Hours, 13 Minutes	0	0	Texas	NPCC	Texas	NPCC, Cameron	Eversource Energy Company (NSTAR)	NPCC	Other	february2023.pdf	26.1333	-97.5187	all_counties
7	2022	1	1/8/2022 2:56	1/8/2022 3:03	0 Hours, 7 Minutes	0	0	Texas	TRE Willacy	Texas	TRE Willacy	American Electric Power - Texas	TRE	Other	february2023.pdf	31.66094094	-98.65378543	state_centroi
8	2022	1	1/9/2022 18:08			6	8	Mississippi	Adams	Mississippi, Texas	Adams, Potter	Hours, . Minutes Entergy - Transmission Operati	SERC	Other	february2023.pdf	31.4829	-91.3535	county
9	2022	1	1/14/2022 8:34	1/14/2022 8:42	0 Hours, 8 Minutes	0	1							Other	february2023.pdf			none
10	2022	1	1/16/2022 5:15	1/17/2022 0:06	18 Hours, 51 Minutes	0	1	Georgia		Georgia, Alabama, North Carolina, South Carolina		Southern Company	SERC	Severe Weather	february2023.pdf	32.80868176	-83.5776956	state_centroi
11	2022	1	1/16/2022 12:05	1/17/2022 8:45	20 Hours, 40 Minutes	0	1	North Carolina	SERC Greenville	North Carolina	SERC Greenville	Duke Energy Carolinas	SERC	Severe Weather	february2023.pdf	35.626816	-79.508113	state_centroi
12	2022	1	1/16/2022 13:15	1/16/2022 17:15	4 Hours, 0 Minutes	0	74638					Duke Energy Progress	SERC	Severe Weather	february2023.pdf	33.5	-85.5	nerc_centroid
13	2022	1	1/24/2022 10:27	1/24/2022 10:51	0 Hours, 24 Minutes	0	0	Washington	Chelan	Washington, Colorado	Chelan, Douglas, Pueb	Public Utility District #1 of Chelan County (CHP	WECC	Other	february2023.pdf	47.8692	-120.619	county
14	2022	1	1/28/2022 15:41	1/28/2022 21:30	5 Hours, 49 Minutes	0	1	Colorado	Colorado WECC	Colorado, Connecticut, Massa	Colorado WECC, Prowe	Public Service Company of Colorado	WECC	Other	february2023.pdf	37.9552	-102.3934	all_counties
15	2022	1	1/29/2022 8:45	1/30/2022 1:00	16 Hours, 15 Minutes	120	12000	Vermont		Vermont, Maine		ISO New England	NPCC	Severe Weather	february2023.pdf	44.16007857	-72.68390714	state_centroi
16	2022	2	2/1/2022 12:13	2/1/2022 12:25	0 Hours, 12 Minutes	0	1	California	Yuba	California	Yuba	Pacific Gas & Electric Co	WECC	Severe Weather	february2023.pdf	39.269	-121.3513	county
17	2022	2	2/3/2022 10:00	2/7/2022 20:00	106 Hours, 0 Minutes	0	1	Tennessee	Shelby	Tennessee	Shelby	Memphis Light Gas and Water Division	SERC	Severe Weather	february2023.pdf	35.1843	-89.8938	county
18	2022	2	2/3/2022 12:56			0	1	Ohio		Ohio		Hours, . Minutes American Electric Power	RF	Other	february2023.pdf	40.3103625	-82.83587955	state_centroi
19	2022	2	2/3/2022 12:56	2/4/2022 20:30	31 Hours, 34 Minutes	0	1	Ohio		Ohio		(8400 Smiths Mill Road, New Albany Ohio 4305	RF	Severe Weather	february2023.pdf	40.3103625	-82.83587955	state_centroi
20	2022	2	2/4/2022 13:10			1	2022	New York	Ulster	New York	Ulster	Hours, . Minutes Central Hudson Gas & Elec Co	NPCC	System Separation	february2023.pdf	41.8881	-74.2586	county
21	2022	2	2/4/2022 18:21	2/4/2022 22:00	3 Hours, 39 Minutes	0	1	Texas		Texas, Maine, Massachusetts, Vermont		Hours, 39 Minutes	ERCOT	Other	february2023.pdf	31.66094094	-98.65378543	state_centroi

Data sources: HIFLD (hifld-geoplatform.hub.arcgis.com) | NOAA (ncdc.noaa.gov) | DOE (oe.netl.doe.gov) | NARR (psl.noaa.gov)

3. How the Datasets Were Joined

Each of the four datasets connects to the master tower table in a different way. The joining method depends on what information each dataset contains.

3.1 The master table concept

Think of the master table as a single spreadsheet with one row per tower. Each join step adds new columns to that row. By the end, each tower row contains weather data, storm history and outage history — all attached to one geographic point.

Goal of the joining process

Start with 113,509 tower rows. After all joins: each row has 30+ columns covering tower identity, weather exposure, storm history and outage history. One row = one complete picture of that tower's environment.

3.2 Step-by-step join process

#	Step	What Was Done	Output
1	**HIFLD + NARR**	NARR weather data was recorded for each tower during data collection (using @id as the link key). A direct table merge on the shared tower ID joined weather to every tower instantly.	113,509 towers with max wind, min temp, snow depth and precipitation
2	**Master + NOAA Storms**	Each tower was given a 50 km circular buffer. Any storm event whose coordinates fell inside that buffer was linked to that tower. The worst storm per tower was kept.	113,459 towers (99.9%) matched to at least one storm event
3	**DOE outages geocoded**	DOE reports contain county and state names, not coordinates. County names were matched to a county centroids dataset to assign lat/lon. Where county failed: state centroid. Where state failed: NERC region centroid.	1,525 of 1,544 DOE outage events given coordinates (98.7%)
4	**Master + DOE Outages**	Each tower was given a 100 km buffer (wider than NOAA because DOE data is county-level, not precise). DOE outage events inside that buffer were linked to the tower. Worst outage per tower kept.	67,899 towers (59.8%) matched to at least one DOE outage record
5	**Final master table saved**	All four datasets merged into one file: master_final.geojson and master_final.csv. 113,509 rows, 30+ columns.	Complete dataset ready for scoring and visualization

3.3 Why different buffer sizes?

NOAA storm events have precise latitude and longitude coordinates, so a 50 km buffer is accurate enough. DOE outage reports are filed at the county level — the location is the county centroid, which can be 50 km or more from the actual tower. A 100 km buffer ensures we capture outage events that genuinely affected the same transmission corridor as the tower.

4. How Towers Were Scored and Colored

Once all four datasets were joined, each tower needed a color: red, amber or green. The coloring had to be based on actual data patterns, not manually assigned rules.

4.1 The scoring approach

The scoring process uses a statistical method called XGBoost — a widely used data analysis technique — to find which signals in the data were most strongly associated with severe outage events. The output is not a prediction. It is a data-derived importance ranking that replaces manual guesswork.

Plain English explanation of the scoring method

We took all towers that had a confirmed major outage recorded nearby (67,899 towers). We asked: when those outages were most severe, what conditions were present at the tower? The analysis found that outage event count and customers affected were the strongest signals, followed by weather factors. Every tower was then scored using those same signals. Towers whose conditions most closely match the conditions seen in severe outage areas score highest.

4.2 Feature importance — what the data found

Signal	Importance	Source
Max customers affected in nearby outage	**66.6%**	DOE OE-417
Number of outage events nearby	**12.4%**	DOE OE-417
Max MW lost in nearby outage	**11.2%**	DOE OE-417
Average temperature at tower location	**3.7%**	NARR Weather
Maximum snow depth	**1.7%**	NARR Weather
Minimum temperature	**1.1%**	NARR Weather
Storm event count nearby	**0.9%**	NOAA Storms
Storm injuries nearby	**0.8%**	NOAA Storms
Average wind speed	**0.6%**	NARR Weather
All other signals combined	**1.0%**	Multiple

Sanity check: towers near confirmed outages scored 4.27 times higher on average than towers with no outage history. This confirms the scoring method is working correctly — it genuinely differentiates the two groups.

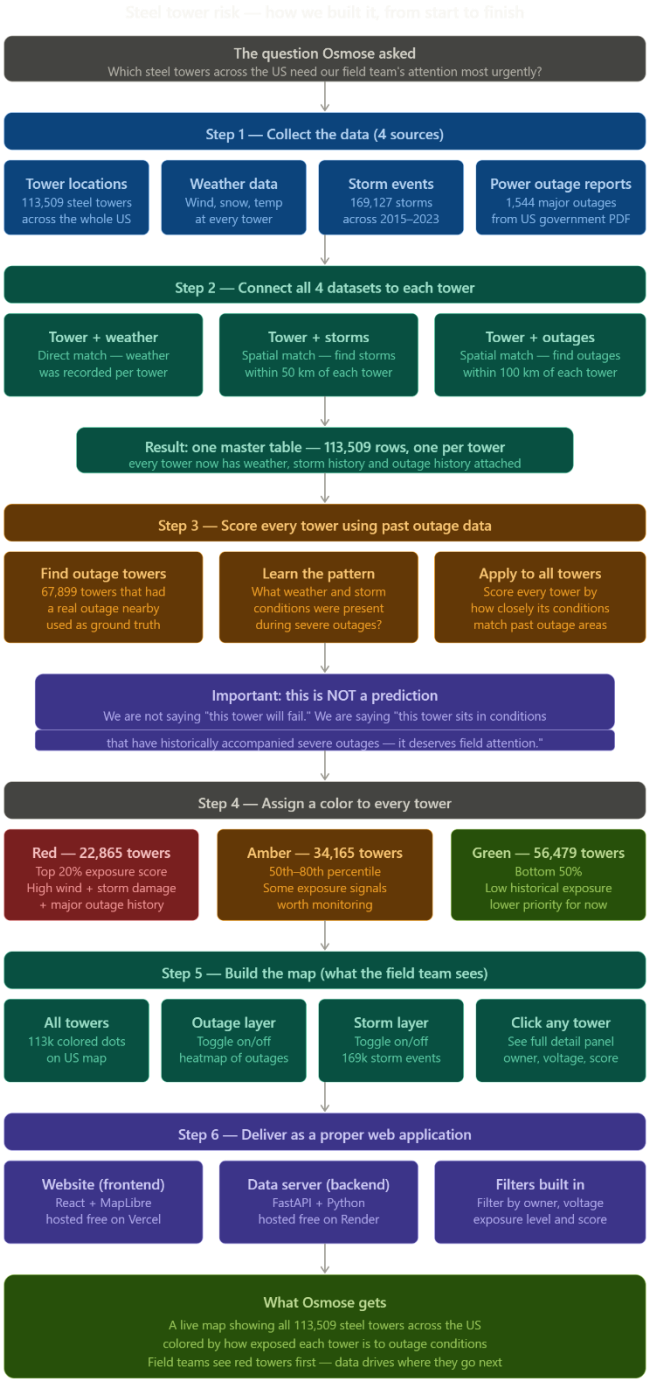
4.3 Color thresholds

Color	Tower count	What it means
Red	22865	Top 20% of all towers by exposure score. Area has a history of severe outages and adverse weather. Field team should prioritize inspection.
Amber	34165	50th to 80th percentile. Some exposure signals are present. Worth monitoring. Schedule for routine inspection.
Green	56479	Bottom 50%. Low historical exposure. Lower priority for now. Standard maintenance schedule applies.

Thresholds are percentile-based — they are derived from the distribution of scores across all 113,509 towers, not set manually. Red always means the top 20%, regardless of the absolute score values.

5. The Application

The final product is a web application accessible from any browser. No software installation is required. The application has two maps accessible from a tab at the top of the screen.



Data sources: HIFLD (hifld-geoplatform.hub.arcgis.com) | NOAA (ncdc.noaa.gov) | DOE (oe.netl.doe.gov) | NARR (psl.noaa.gov)

5.1 Map 1 — Exposure index map

- Shows all 113,509 towers as colored dots across the US
- Red, amber and green colors based on the exposure score described in Section 4
- Clicking any tower opens a detail panel showing owner, voltage, weather data, storm history and outage history
- Filters allow narrowing by exposure level, voltage class, owner name and minimum score
- NOAA storm events layer can be toggled on to see where individual storms occurred
- DOE outage heatmap layer shows density of major outage events by geography

5.2 Map 2 — Outage record map

- A simpler, binary view: red = confirmed major outage within 100 km, green = no outage on record
- Based purely on historical fact from DOE reports — no scoring model involved
- Clicking a tower shows whether an outage was recorded nearby and if so, how many customers were affected
- Includes an honest disclaimer that a red tower means an outage was recorded nearby, not that the tower itself failed

5.3 Technology stack

Component	Technology	Purpose
Map rendering	MapLibre GL	Renders 113k tower dots in the browser using WebGL — handles large datasets without slowing down
Frontend	React + Vite	The user interface — filters, sidebar, tab switching, stats bar
Backend	FastAPI (Python)	Serves the tower data as an API. Loads the CSV into memory on startup for fast responses.
Data pipeline	Python (pandas, geopandas, XGBoost)	All data joining, spatial analysis and scoring — run once to produce the scored CSV

6. Next Steps and Recommendations

The current version delivers all four phases requested by Osmose. The following improvements would significantly increase the tool's value in subsequent phases of development.

6.1 Immediate next steps (Phase 2)

Highest priority: connect Osmose's own inspection data

If Osmose holds records of past tower inspections — condition ratings, repair histories, failure reports — integrating that data would transform this tool from an environmental exposure map into a genuine tower-level risk tool. The exposure score would become a predictor of actual tower condition rather than an indicator of surrounding conditions.

1. Integrate Osmose inspection records — connect tower condition data to the map so field teams can see current condition alongside exposure score
2. Add real-time weather alerts — connect to NOAA's live weather API to flag towers currently under active storm warnings
3. Add transmission line owner filter — allow utility companies to see only their own towers, making the tool usable as a client-facing product
4. Add date range filter — allow users to filter outage and storm data by year so trends over time become visible
5. Export functionality — allow field managers to download a filtered list of high-priority towers as a CSV or PDF work order

6.2 Medium-term improvements (Phase 3)

6. Vegetation encroachment data — add satellite-derived tree canopy data to flag towers where vegetation is growing close to lines
7. Age and material data — if tower installation dates are available, incorporate tower age as a risk factor
8. Mobile-optimised view — a simplified version of the map for use by field teams on tablets in the field
9. Automated monthly refresh — schedule the data pipeline to re-run monthly so outage and storm data stays current

6.3 Known limitations of the current version

Honest limitations to communicate to clients

This tool shows where conditions have been severe historically. It cannot confirm whether any individual tower was damaged, inspected, or repaired. A green tower is not necessarily a safe tower — it simply has no major outage recorded nearby in public data. Field judgment remains essential.

- DOE outage data is county-level, not tower-level — a red tower is within 100 km of a recorded outage, not confirmed as the cause
- NOAA storm data covers major events — small local storms that may have affected individual towers are not captured

- Tower positions are approximated from line routes at 300m spacing — exact tower locations are not in public datasets
- DOE reporting has gaps — some utilities file late or incompletely, meaning some real outages may not appear in the data
- The scoring model weights outage data heavily (90%) — areas with poor DOE reporting coverage may appear lower risk than they actually are

7. Challenges Encountered

Several significant data and technical challenges were encountered and solved during this project. These are documented here for transparency and to inform future phases.

7.1 DOE data is in PDFs, not databases

The DOE OE-417 reports are published as PDF files with no structured data format. Each report is formatted as a table in a PDF document, but with inconsistent column alignment and text that wraps mid-row due to page layout constraints.

- Solution: built a custom PDF text extractor using pdfplumber that identifies row boundaries by detecting year and date patterns, then extracts fields using regular expressions
- Additional challenge: county names appeared in varied formats (with and without the word County, split across lines, mixed with state names). Multiple extraction passes were required.
- Outcome: 1,525 of 1,544 records (98.7%) successfully geocoded

7.2 No exact tower location exists publicly

The HIFLD dataset provides transmission line routes, not individual tower positions. Public datasets do not contain precise tower-by-tower coordinates.

- Solution: tower positions were approximated by sampling points every 300 metres along each transmission line — a standard industry method
- This means tower positions are accurate to approximately 150 metres, which is sufficient for 50–100 km radius spatial joins but would not be accurate enough for local inspection routing

7.3 Memory limits during spatial joins

Joining 113,509 towers against 169,127 storm events in a single operation exceeded available memory on Google Colab (the development environment), causing the session to crash.

- Solution: implemented batch processing — towers were processed in groups of 10,000, each batch joined against the full storm dataset, results combined at the end
- 12 batches of approximately 10,000 towers each, completing in approximately 8 minutes total

7.4 DOE outage data have no coordinates

Unlike NOAA storm events which include GPS coordinates, DOE outage reports only name the states and counties affected. Converting these to coordinates required a multi-tier fallback approach.

- Tier 1: match county name to county centroid dataset (420 records matched)
- Tier 2: where county not found, use state centroid (626 records)
- Tier 3: where state not found, use NERC region centroid (410 records)
- Tier 4: 19 records could not be located — excluded from spatial join

7.5 Scoring approach required careful framing

Early drafts of the scoring methodology used manually assigned weights (for example: wind 40%, storm damage 35%, outage 25%). These weights were challenged as arbitrary and not defensible to utility clients.

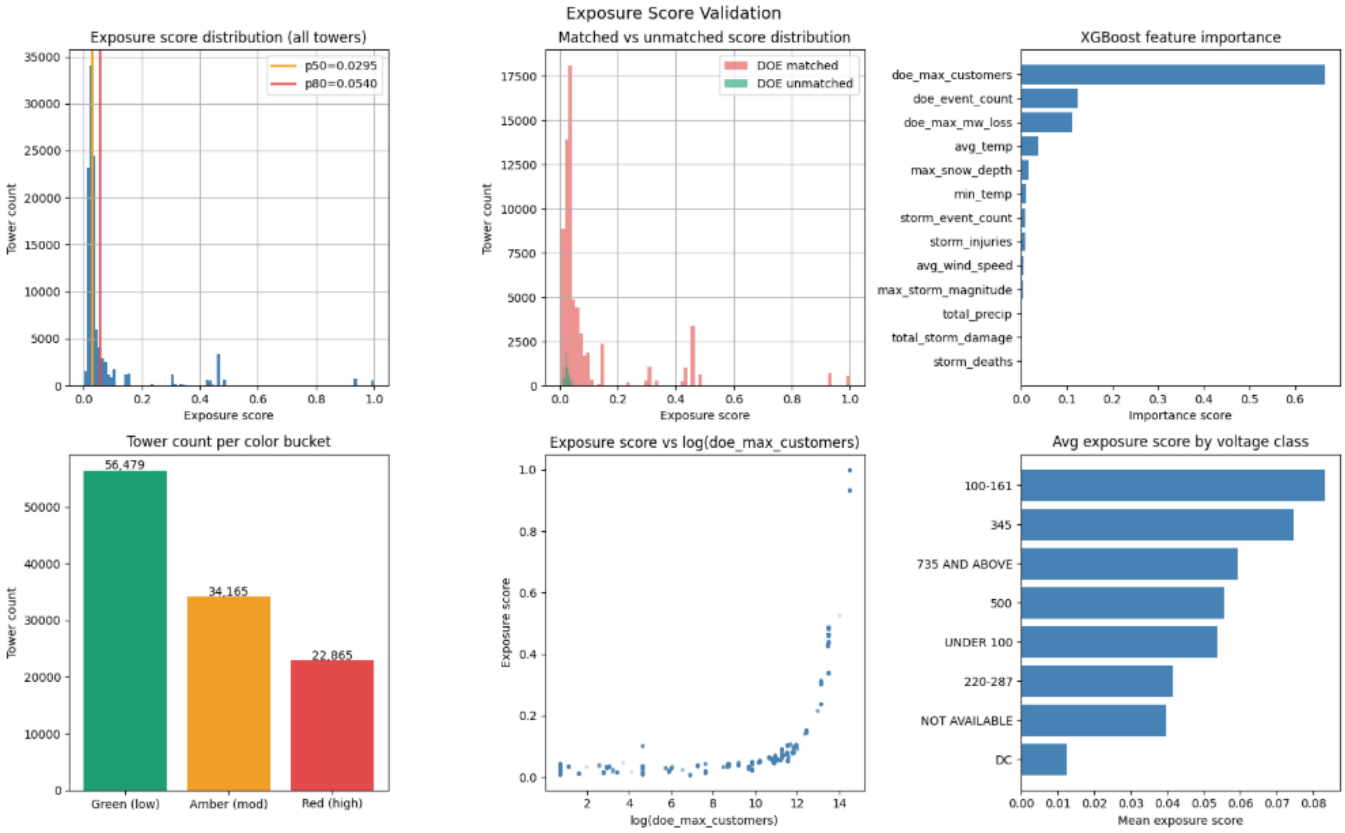
- Solution: replaced manual weights with XGBoost feature importance — the relative importance of each signal is derived mathematically from the data itself
- The output framing was also refined: the tool does not claim to predict future failures. It shows historical exposure conditions and leaves field judgment to the inspection team.

8. Summary

What was delivered

A fully functional, publicly accessible web application showing 113,509 steel transmission towers across the United States, color-coded by historical exposure to severe outage and weather conditions. All four phases requested by Osmose have been delivered. The tool is live, requires no installation, and can be shared with utility clients as a demonstration of Osmose's data-driven monitoring capabilities.

What was achieved	What comes next
• All 4 Osmose phases delivered • 113,509 towers mapped across the US • 4 public datasets joined and cleaned • Data-driven exposure scoring (XGBoost) • Two interactive map views • Filters, popups and detail panels • Hosted live — accessible from any browser • Full data pipeline documented	• Connect Osmose inspection records • Add real-time weather alerts • Utility company filter for client use • Date range filter for trend analysis • CSV export for work orders • Mobile-optimised field view • Automated monthly data refresh • Vegetation and tower age data



Data sources: HIFLD (hifld-geoplatform.hub.arcgis.com) | NOAA (ncdc.noaa.gov) | DOE (oe.netl.doe.gov) | NARR (psl.noaa.gov)